

The configuration the extrapolated anchor chooses: full results of the inference-configuration sweep

17 August 2026. Last week's multi-day sweep scored 648 combinations of model state and sampling depth across eight research Reynolds numbers, using only the spectral anchor extrapolated from coarse observations, and audited 96 of those combinations against held-out ground truth. This report mines that sweep with information that did not exist when it ran - the measured per-regime anchors built since - and answers three questions: which configuration the extrapolated anchor alone selects at every regime, how accurate that anchor actually was, and whether the structural guard in the selection rule earned its place. The selection replay used here reproduces all sixteen recorded picks of the original run exactly before any variant is computed, so every counterfactual below shares the original's exact semantics.

1. What was swept and how selection works

One lineage fact first, because the tables below are misread without it: EVERY model in the sweep is the same Reynolds-1000-pretrained diffusion model. The candidates differ only in what the reinforcement-learning fine-tune targeted: nothing (the base), the Reynolds-1000 anchor (two reference checkpoints, transported to every regime), or - at Reynolds 2000 only, where such fine-tunes exist - the Reynolds-2000 anchor (twelve checkpoints). So 'the Re=2000 model' is not a differently pretrained network; it is the same backbone whose fine-tune was aimed at the target regime.

Each candidate was run at nine sampling depths, from a single shallow pass to a seven-pass cascade. Every combination was scored blind: the ratio of its output energy to the extrapolated anchor's over wavenumbers 10 to 95, on the anchor's own source sequences. The rule then shortlists cells whose blind score falls in the engraved band, discards those whose fine-scale structure has drifted far from the base reconstruction's (the guard), and takes the survivor nearest the set-point, ties resolved toward shallower depth and less training. Where no cell survives, the fallback takes the cell nearest the set-point regardless of band. Afterwards - and only afterwards - graded retention against ground truth was measured on held-out sequences for the audited subset.

2. The answer: what the extrapolated anchor alone chooses

The table below is the centrepiece: at every regime, the configuration selected using nothing but the extrapolated anchor's blind score (no guard), and its graded quality. Where the two calibration arms of the original ablation disagree, both are shown.

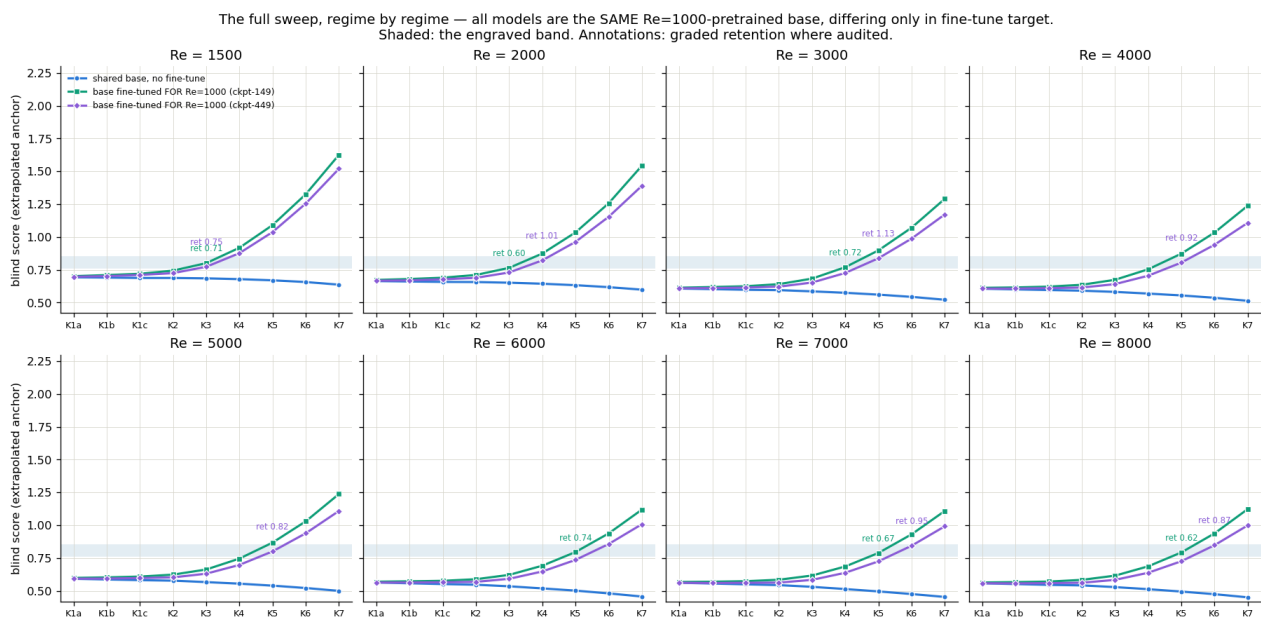
Regime	Chosen by the extrapolated anchor alone	blind	graded retention
1500	three-pass cascade (86 steps) base fine-tuned for Re=1000, checkpoint 149 (transported)	0.801	0.715
2000	three-pass cascade (86 steps) base fine-tuned for Re=2000, checkpoint 149	0.789	0.817
3000	single-regime arm: four-pass cascade (110 steps) base fine-tuned for Re=1000, checkpoint 149 (transported)	0.770	0.716
	both-regimes arm: five-pass cascade (140 steps) base fine-tuned for Re=1000, checkpoint 449 (transported)	0.838	1.133
4000	five-pass cascade (140 steps) base fine-tuned for Re=1000, checkpoint 449 (transported)	0.805	0.921
5000	five-pass cascade (140 steps) base fine-tuned for Re=1000, checkpoint 449 (transported)	0.801	0.821
6000	five-pass cascade (140 steps) base fine-tuned for Re=1000, checkpoint 149 (transported)	0.796	0.736

7000	five-pass cascade (140 steps) base fine-tuned for Re=1000, checkpoint 149 (transported)	0.789	0.672
8000	five-pass cascade (140 steps) base fine-tuned for Re=1000, checkpoint 149 (transported)	0.793	0.621

In plain terms: near the training regime the anchor chooses a three-pass cascade; by Reynolds 4000 it has moved to the five-pass cascade and stays there through 8000, switching from the later transported checkpoint (449) to the earlier, more conservative one (149) as the regime grows more remote. At Reynolds 2000, where regime-targeted fine-tunes competed, it passes over the transported models for the early regime-targeted checkpoint. Graded quality of these blind choices: 0.72 to 0.92 retention through Reynolds 5000, decaying to 0.62 at 8000 - two to four times what the un-fine-tuned base achieves anywhere.

3. The full grid, regime by regime

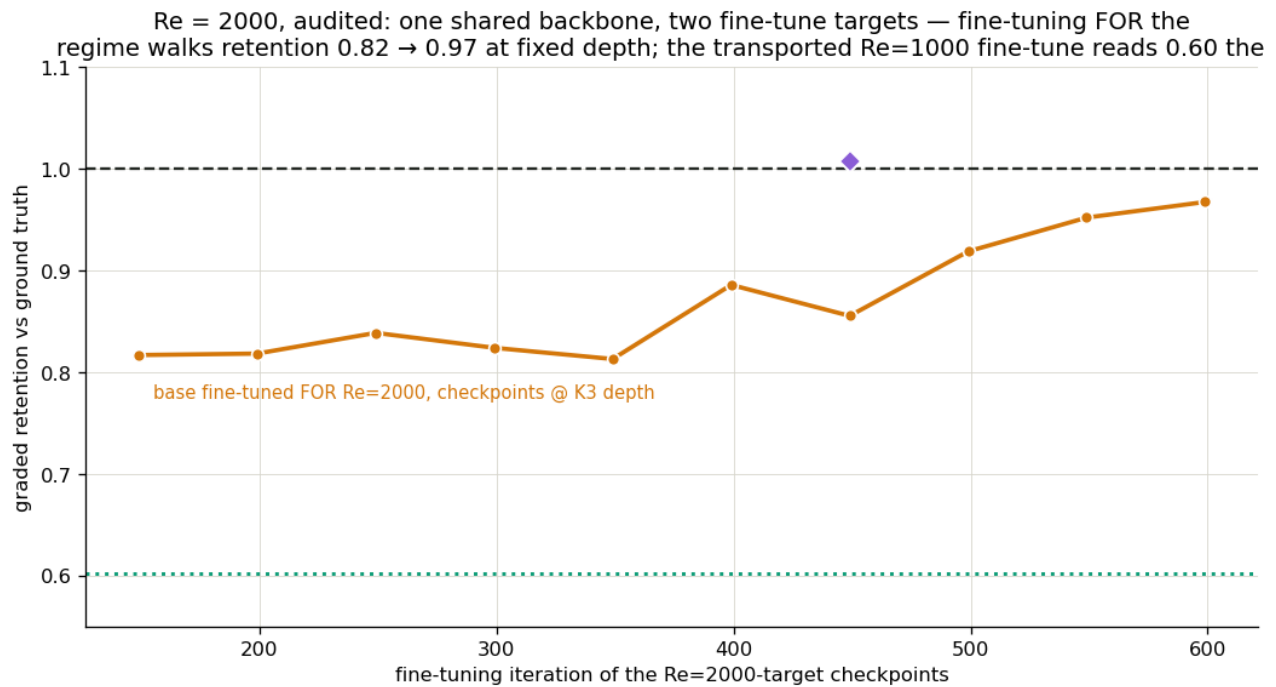
Below, every cell of the sweep at the seven regimes where three candidates competed: the blind-score ladders across all nine depths, with graded retention annotated wherever a cell was audited. Two structural facts are visible everywhere. The fine-tuned curves rise steeply with depth while the base's drift down - depth multiplies the fine-tune's learned energy dose, and the band (shaded) intersects the rising curves exactly once per model, which is why selection is stable. And the two transported checkpoints bracket the band at nearly every regime, so the anchor always has a candidate to land on.



All 27 cells per regime (three models, nine depths). K1a/b/c are the three single-pass variants; K2 through K7 the cascades.

4. Reynolds 2000 in full: fine-tuned FOR the regime versus transported

Reynolds 2000 is the one regime where the sweep contains fine-tunes aimed at the target regime itself, so it is the only place the question 'is a transported fine-tune as good as a regime-targeted one?' has an answer. It is not close. At the three-pass depth the audited regime-targeted checkpoints read retention 0.82 to 0.97 - improving steadily with fine-tuning iteration - while the transported Reynolds-1000 checkpoint at the very same depth reads 0.60. The transported model only reaches comparable numbers one depth level deeper (checkpoint 449 at the four-pass cascade grades 1.007, the best audited retention in the grid, though with visibly worse fine-scale placement: 0.77 against the regime-targeted checkpoints' 0.84). Same backbone in every case - the difference is entirely which anchor the fine-tune pursued, plus the depth compensating for the transported model's calibrated-for-Re=1000 dose.



Audited retention at Reynolds 2000. The rule's pick (checkpoint 149 at three-pass depth, 0.817) is conservative: it prefers the placement-safer regime-targeted cell over the transported model's deeper, hotter cell.

The full blind grid at Reynolds 2000 (fifteen models, nine depths) in numbers, graded retention in parentheses where audited:

model	K1a	K1b	K1c	K2	K3	K4	K5	K6	K7
base	0.66 (0.28)	0.66 (0.28)	0.66 (0.28)	0.66 (0.29)	0.65 (0.31)	0.64 (0.32)	0.63 (0.33)	0.62 (0.33)	0.60 (0.31)
r1k-149	0.67	0.68	0.69	0.71	0.76 (0.60)	0.87	1.03	1.26	1.54
r1k-449	0.67	0.67	0.68	0.69	0.73	0.82 (1.01)	0.96	1.15	1.39
0049	0.67	0.68	0.68	0.70	0.73	0.80 (0.92)	0.89	1.00	1.13
0099	0.67	0.68	0.69	0.71	0.76	0.87	1.01	1.20	1.43
0149	0.68	0.69	0.70	0.72	0.79 (0.82)	0.92	1.10	1.34	1.62
0199	0.68	0.68	0.69	0.72	0.78 (0.82)	0.90	1.08	1.29	1.54
0249	0.67	0.68	0.69	0.72	0.79 (0.84)	0.93	1.14	1.40	1.70
0299	0.67	0.68	0.70	0.72	0.80 (0.82)	0.94	1.16	1.44	1.78
0349	0.68	0.68	0.70	0.72	0.79 (0.81)	0.93	1.14	1.41	1.74
0399	0.68	0.69	0.70	0.73	0.81 (0.89)	0.97	1.21	1.53	1.90
0449	0.67	0.68	0.69	0.71	0.77 (0.86)	0.91	1.11	1.37	1.68
0499	0.67	0.68	0.70	0.72	0.80 (0.92)	0.97	1.22	1.54	1.91
0549	0.67	0.69	0.70	0.73	0.82 (0.95)	1.01	1.29	1.65	2.09

0599	0.67	0.68	0.70	0.73	0.82 (0.97)	1.01	1.30	1.67	2.10
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Rows 2-3 are the transported Reynolds-1000 fine-tune; rows 4 onward the Reynolds-2000-targeted checkpoints by fine-tuning iteration. All share one pretrained backbone.

5. The guard almost never mattered - the labels did

The original run reported 'no cell in band' at every regime except Reynolds 2000, which read as the guard vetoing everything. The faithful replay resolves this: with the guard removed, the selected cell is IDENTICAL at fifteen of the sixteen regime-and-arm combinations. The guard's vetoes changed the label (in-band pick versus fallback), but the fallback landed on the same cell the blind score would have chosen anyway. The single exception is Reynolds 1500 in the single-regime arm, where the guarded rule's fallback chose checkpoint 449 (graded 0.745) while the pure blind rule chooses checkpoint 149 (graded 0.715) - a 0.03 difference, in the guard's favour. Conclusion: the guard is close to a no-op in outcomes and its reference drifts across regimes; a revised rule can drop it or re-specify it without loss.

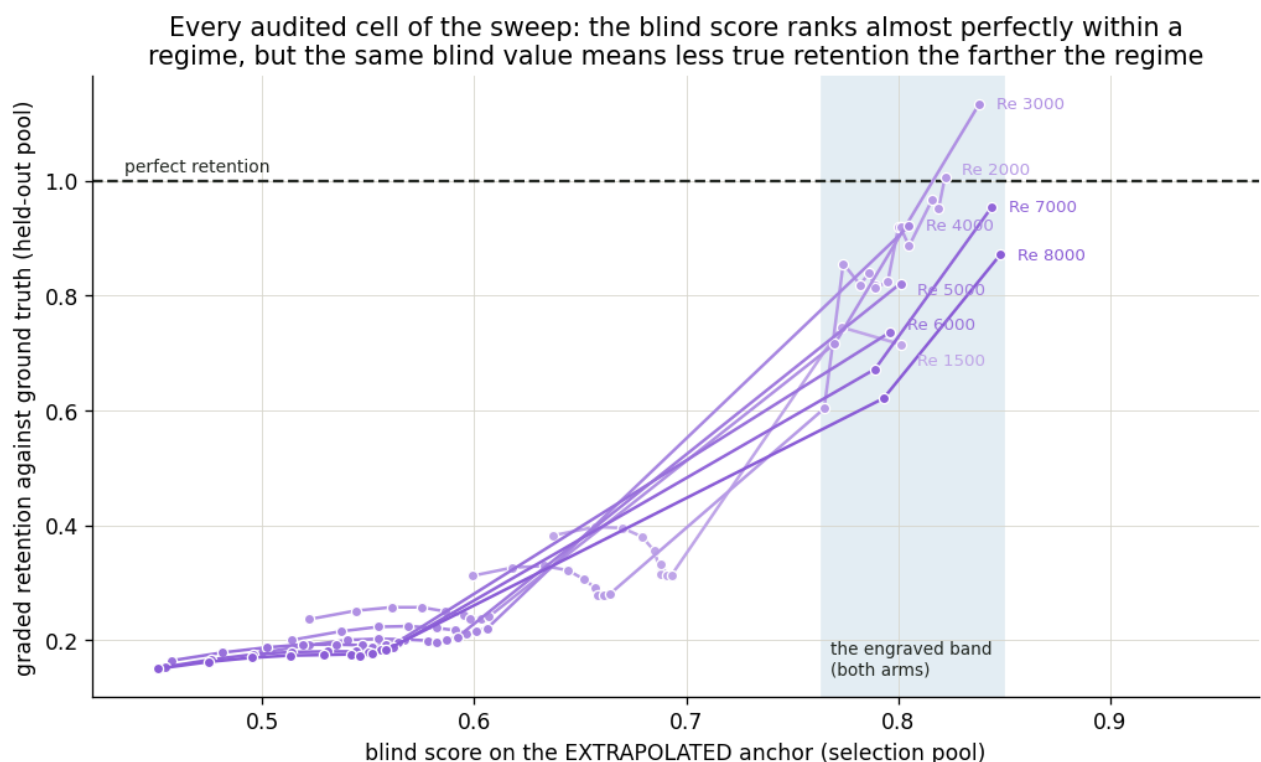
6. How accurate was the extrapolated anchor, exactly

Comparing the extrapolated anchor against the measured anchor (built later from training-sequence ground truth) band by band shows the anchor's error is a SHAPE error, not a level error: it is too energetic in the middle wavenumbers and too weak in the tail, at every regime.

Regime	blind band [10,96): extrapolated / measured	reward band [32,96)
1500	1.074 (hot)	0.934 (cold)
2000	1.109 (hot)	0.935 (cold)
3000	1.186 (hot)	0.964 (cold)
4000	1.218 (hot)	0.964 (cold)
5000	1.125 (hot)	0.885 (cold)
6000	1.221 (hot)	0.949 (cold)
7000	1.183 (hot)	0.887 (cold)
8000	1.141 (hot)	0.848 (cold)

Hot by 7 to 22 percent where the blind score integrates, cold by 4 to 15 percent where the reward concentrates - the 'too narrow in k' diagnosis of the earlier audit, now quantified per regime. These ratios are directly usable as a correction curve for a second-generation anchor.

7. Despite that, the blind score was nearly optimally calibrated



All 96 audited cells: blind score (horizontal) against graded retention (vertical), one line per regime.

Two facts. Within any single regime the blind score ranks configurations almost perfectly (correlation 0.85 to 0.97; pooled 0.93): whatever the anchor's absolute error, MORE blind energy reliably means MORE true energy. And at the two regimes where audited models actually cross perfect retention, the blind value corresponding to retention exactly 1 is 0.821 (Reynolds 2000) and 0.816 (Reynolds 3000) - the engraved set-point of 0.81 was within one percent of optimal, twice, independently. What does NOT transfer is the absolute meaning of a blind value across regimes: inside the engraved band, true retention fans out from 1.13 at Reynolds 3000 to 0.62 at 8000. Most of that fan-out is model reach, not anchor error - at the far regimes the audited models never cross

retention 1 at ANY depth, so no blind value could map to perfection there.

8. What this supports going forward

(1) The engraved blind machinery stands: selection from the extrapolated anchor alone reproduces the full rule's choices at 15 of 16 arm-regime combinations and needed neither the guard nor the second calibration regime. (2) A second-generation anchor should correct the k-shape using the per-regime hot/cold ratios of section 4; against corrected anchors the set-point re-expresses to roughly 0.93. (3) The far-regime ceiling is a model limitation, not a selection limitation - the transported $Re=1000$ fine-tune simply runs out of reach past Reynolds 5000, which is the same conclusion the conditioning arc reached from the other direction. Provenance: sweep `src/ddpo_ft/setpoint_ablation.py` (results `base_results/setpoint_ablation.npz`); replay `src/ddpo_ft/ablation_choice_replay.py` (validated against the run log); anchor comparison `src/ddpo_ft/ablation_mining.py`.